Assignment for AI Developer

1.Chatbot development (Use RAG)

1. **Develop a Chatbot as per the below objectives** :
    - Build a chatbot using a GPT-based model (e.g., GPT-4 or similar).
    - Enable the chatbot to read and comprehend content from a given PDF file (Use the PDF attached in the email)
    - Allow users to interact with the chatbot by asking questions based on the content of the PDF.
2. **API Development**:
    - Expose the chatbot functionality as an API endpoint.
    - Ensure that the API accepts user queries as input and returns responses generated by the GPT model.
3. **Validation**:
    - Validate the chatbot API functionality using **Postman**.

What is Expected

- A Python-based solution (preferably using **Flask** or **FastAPI**) to build the API.
- Utilize a library like **PyPDF2** or **pdfplumber** to extract text from the PDF.
- Fine-tune or prompt the GPT model to answer questions accurately based on the extracted content.
- A well-documented **Postman collection** for testing the API with sample queries.

Deliverables

1. **Codebase**:
    - Python scripts with clear comments and modular structure.
    - Requirements file (requirements.txt) listing dependencies.
2. **Postman Collection**:
    - A collection containing pre-configured API requests (e.g., for uploading PDF, sending user queries, and receiving responses).
3. **Sample PDF**:
    - A sample PDF used for testing and demonstration.
4. **Output Screens**:

Attach screenshots of the following:

- API request and response in Postman.
- Terminal/console logs during execution.
- Chatbot response to a user question.

Output Screens to Attach

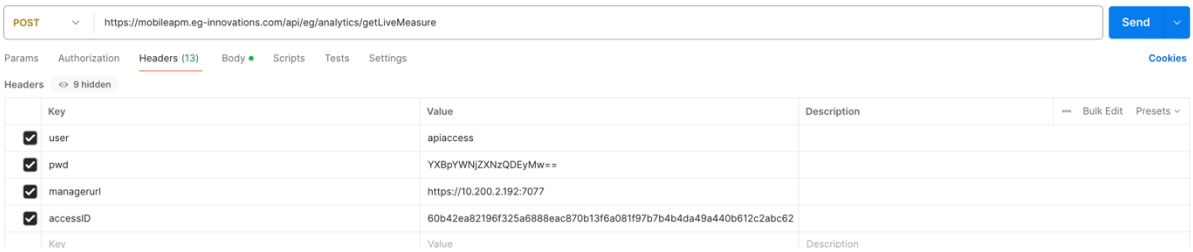
1. **Postman Output**:
   - Screenshot of a user query sent via Postman and the corresponding chatbot response.
2. **Console Logs**:
   - Screenshot showing logs for loading the PDF, extracting text, and processing the GPT response.
3. **API Demo**:
   - Screenshot of chatbot responding to a user question based on the provided PDF content.

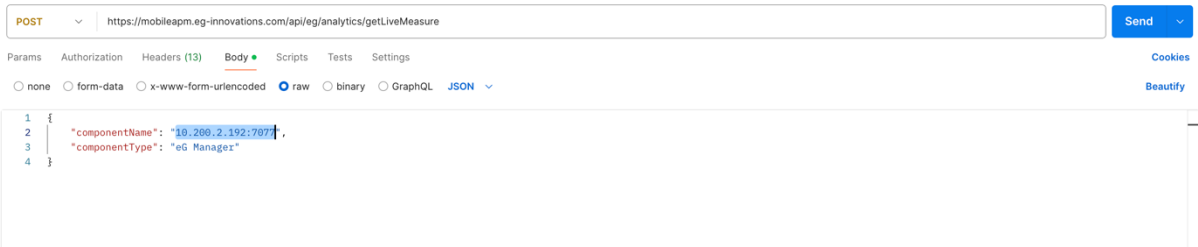
Questions to ask to the BOT :

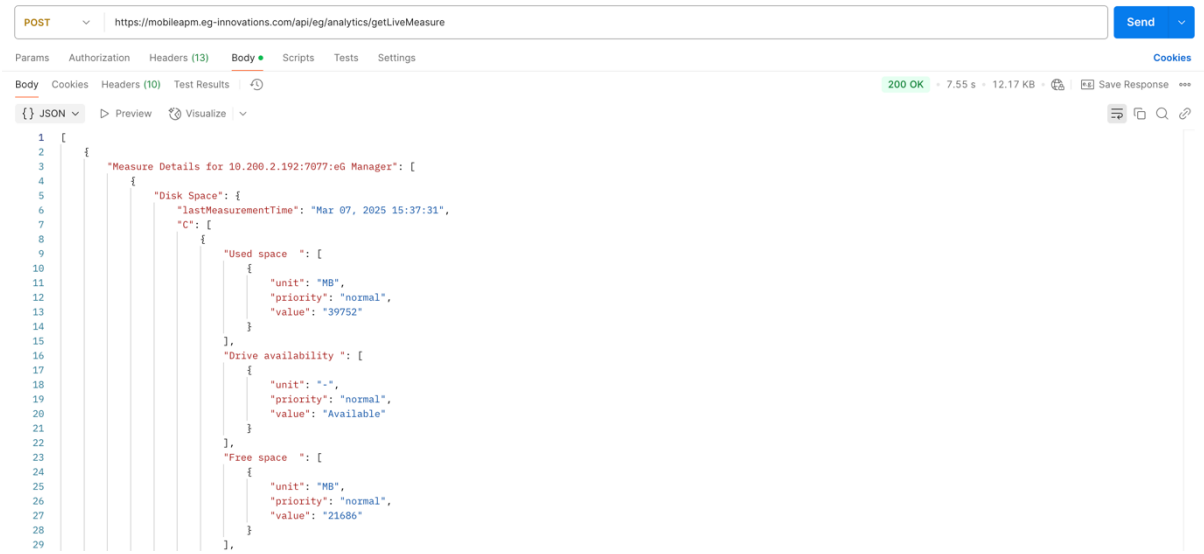
1. Do you support WebLogic and what versions of WebLogic is supported
2. Do you support IBM Mainframe?
3. Do you support Oracle Database monitoring and what versions supported?
4. Can you share the MSSQL server versions supported by eG Enterprise?
5. Does eG Enterprise supports 'Open Telemetry' ?

2. Chatbot development (Use RAG)

4. **Develop a Chatbot as per the below objectives** :
   - Build a chatbot using an offline model (Hugging face or Ollama).
   - Enable the chatbot to read data from the API ; Attached the API collection ; You could use the same end point and credentials set with the Headers and the params sent from the body ; Access the API using POSTMAN to ensure that its giving data







- Allow users to interact with the chatbot by asking questions based on the API response

5. **API Development**:

   - Expose the chatbot functionality as an API endpoint.
   - Ensure that the API accepts user queries as input and returns responses generated by the offline model.

6. **Validation**:

   - Validate the chatbot API functionality using **Postman**.

What is Expected

- A Python-based solution (preferably using **Flask** or **FastAPI**) to build the API.
- Fine-tune or prompt the GPT model to answer questions accurately based on the extracted content.
- A well-documented **Postman collection** for testing the API with sample queries.

Deliverables

5. **Codebase**:
    - Python scripts with clear comments and modular structure.
    - Requirements file (requirements.txt) listing dependencies.
6. **Postman Collection**:
    - A collection containing pre-configured API requests (e.g., for uploading PDF, sending user queries, and receiving responses).
7. **Sample API** :
    - A sample API collection is attached .
8. **Output Screens**:

    Attach screenshots of the following:

    - API request and response in Postman.
    - Terminal/console logs during execution.
    - Chatbot response to a user question.

Output Screens to Attach

4. **Postman Output**:
    - Screenshot of a user query sent via Postman and the corresponding chatbot response.
5. **Console Logs**:
    - Screenshot showing logs for loading the API , extracting response , and processing the offline model response.

Questions to ask to the BOT :

1. What is the free space for the server 10.200.2.192 ?

2. What is the CPU Utilization for the server 10.200.2.192?

3. What is the memory usage for the server 10.200.2.192?